

NETWORK SECURITY

VPN, CISCO IOS & VLAN

SECURE CONNECTION. SMARTER NETWORKS. STRONGER DEFENSE.

- 
CONFIDENTIALITY
- 
INTEGRITY
- 
AUTHENTICATION
- 
PROTECTION



```
Cisco IOSv
User Access Verification
Password: *****
Router# show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0 192.168.10.1  YES manual  up      up
Router#
```



- 
VPN
- 
VLAN
- 
CISCO IOS
- 
SECURITY

@manishburdak45



AGENDA

TOPICS OVERVIEW

“

Secure Connection.
Smarter Networks.
Stronger Defense.



”

- | | | | | | |
|----|--|---------------------------|----|--|--------------------------------------|
| 01 | | Introduction to VPN | 16 | | SSH vs Telnet |
| 02 | | How VPN Works | 17 | | VLAN Introduction |
| 03 | | Real-Life VPN Scenario | 18 | | VLAN Architecture Diagram |
| 04 | | VPN Components | 19 | | VLAN IDs & VLAN Types |
| 05 | | VPN Ports & Protocols | 20 | | Access vs Trunk Ports |
| 06 | | IPsec Introduction | 21 | | IEEE 802.1Q VLAN Tagging |
| 07 | | AH vs ESP | 22 | | VLAN Configuration (Linux / Windows) |
| 08 | | Transport vs Tunnel Mode | 23 | | Wireshark VLAN Analysis |
| 09 | | IKE & Secure Key Exchange | 24 | | VLAN Hopping Attack |
| 10 | | PPTP & Weaknesses | 25 | | Double Tagging Attack |
| 11 | | VPN Security Attacks | 26 | | VXLAN Introduction |
| 12 | | VPN in Cybersecurity | 27 | | CDP (Cisco Discovery Protocol) |
| 13 | | VPN Interview Questions | 28 | | STP (Spanning Tree Protocol) |
| 14 | | Cisco IOS Introduction | 29 | | Tools for Analysis & Testing |
| 15 | | Cisco Password Types | 30 | | Final Summary & Interview Questions |



INTRODUCTION TO VPN

“

VPN creates a **secure encrypted tunnel** over the internet allowing remote users to access private networks safely.



”



WHAT IS VPN?

A VPN (Virtual Private Network) is a technology that creates a **secure, encrypted connection (tunnel)** between a remote device and a private network over the internet.



WHY VPN IS USED?

- ✓ Secure remote access to private resources
- ✓ Data encryption protects from eavesdropping
- ✓ Access internal network from anywhere
- ✓ Cost-effective compared to leased lines
- ✓ Connect multiple sites securely (Site-to-Site VPN)



KEY BENEFITS



Confidentiality



Remote Access

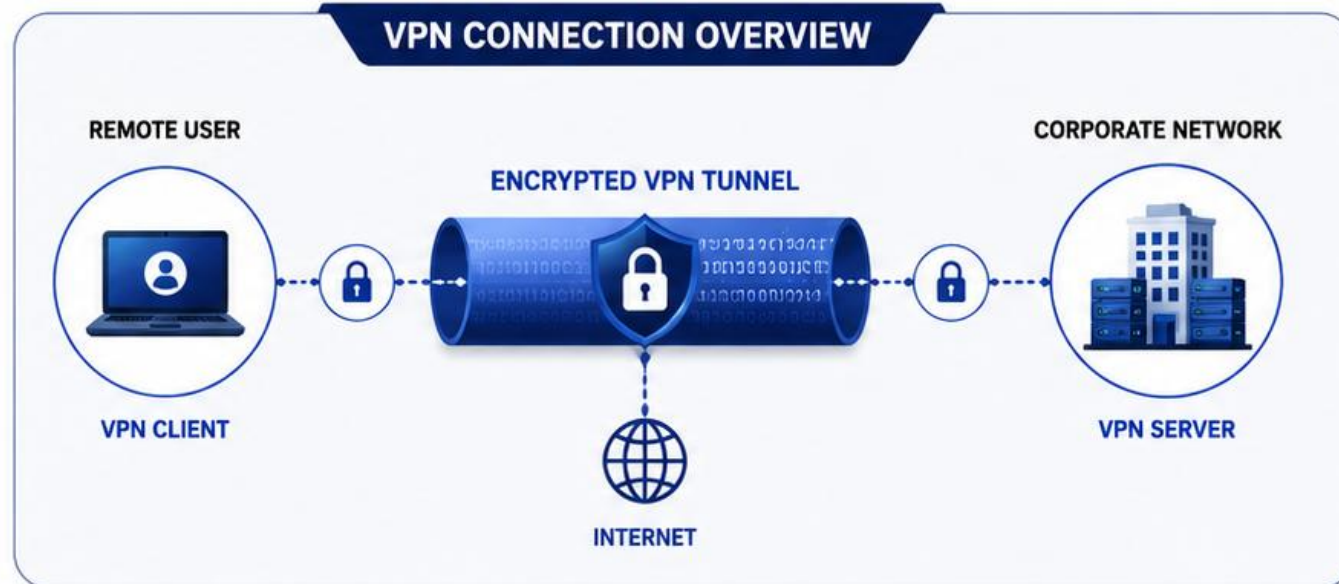


Data Integrity



Cost Effective

VPN CONNECTION OVERVIEW



COMMON VPN PROTOCOLS & PORTS

PROTOCOL	PORT	TRANSPORT	DESCRIPTION
PPTP	TCP 1723	TCP	Point-to-Point Tunneling Protocol
IKE / IKEv1 / IKEv2	UDP 500	UDP	Internet Key Exchange
IPsec (ESP)	UDP 4500	UDP	Encapsulating Security Payload



IMPORTANT

VPN encrypts data and authenticates users, ensuring secure and confidential communication over untrusted networks like the internet.



VPN

VIRTUAL PRIVATE NETWORK

HOW VPN WORKS

VPN works by creating an **encrypted tunnel** between the user and the destination network over the internet.

“

Your data is encrypted, tunneled, and securely delivered to the destination network.



”

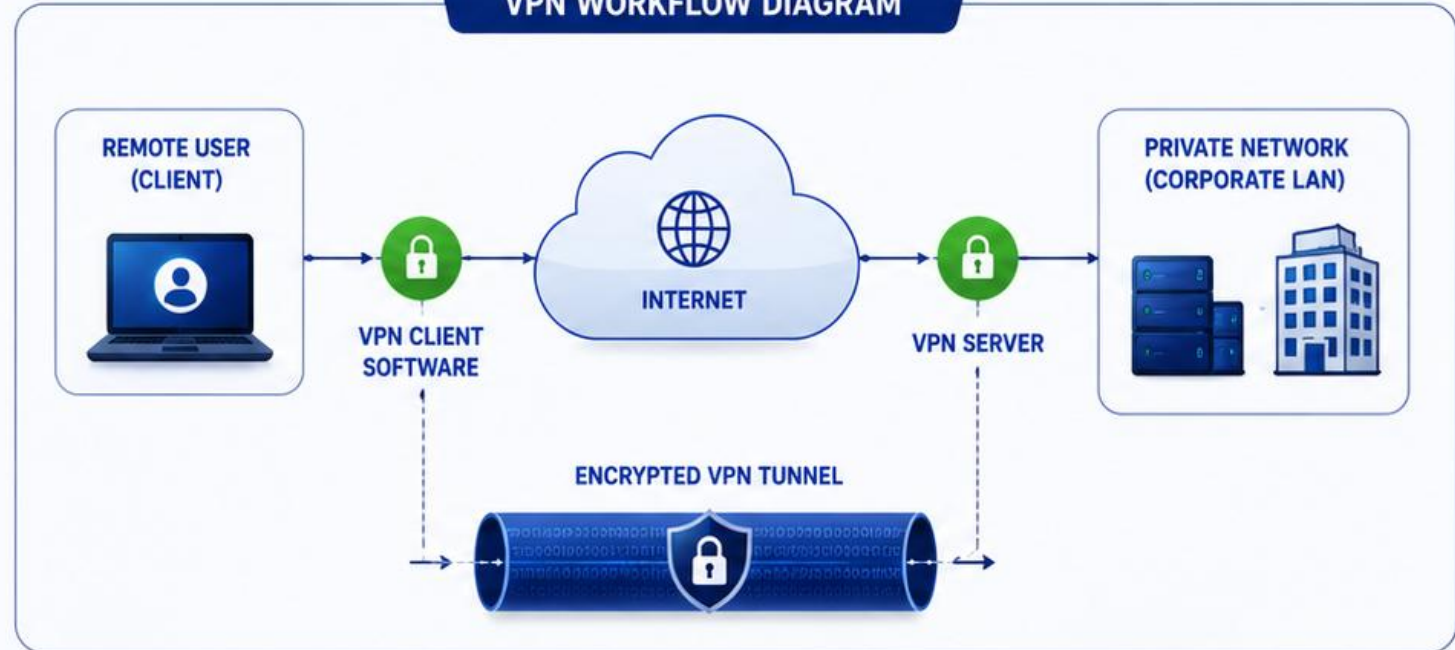
03

/ 30

STEP-BY-STEP PROCESS

- 01 User Request**
The user initiates a connection to the VPN server.
- 02 Authentication**
The VPN server authenticates the user.
- 03 Encrypted Tunnel Creation**
An encrypted tunnel is created over the internet.
- 04 Secure Data Transmission**
All data travels through the encrypted tunnel.
- 05 Access to Private Network**
User can securely access resources in the private network.

VPN WORKFLOW DIAGRAM



KEY BENEFITS



Data Security



Remote Access



Access from Anywhere



Protects Private Information



Cost Effective



Secure Site-to-Site Connection



VPN

VIRTUAL PRIVATE NETWORK



REAL-LIFE VPN SCENARIO

VPN allows remote users to **securely** access their organization's **internal network** and resources over the internet.

“ Connect Securely.
Work Anywhere.
Stay Protected.



04
/ 30

SCENARIO



1. Remote Employee

An employee works from home and needs to access company files and applications.



2. VPN Connection

The employee connects to the company VPN using the internet.



3. Secure Tunnel

An encrypted tunnel is established between the employee and the company network.



4. Access Granted

The employee securely accesses internal resources as if they are in the office.

VPN CONNECTION DIAGRAM

REMOTE USER



Employee at Home



ENCRYPTED VPN TUNNEL



SECURE & ENCRYPTED
DATA TRANSFER

COMPANY NETWORK



File Server Database Application

BENEFITS



Secure Access



Access from
Anywhere



Data Encryption



Cost Effective



Increased
Productivity



Safe Remote
Work Environment



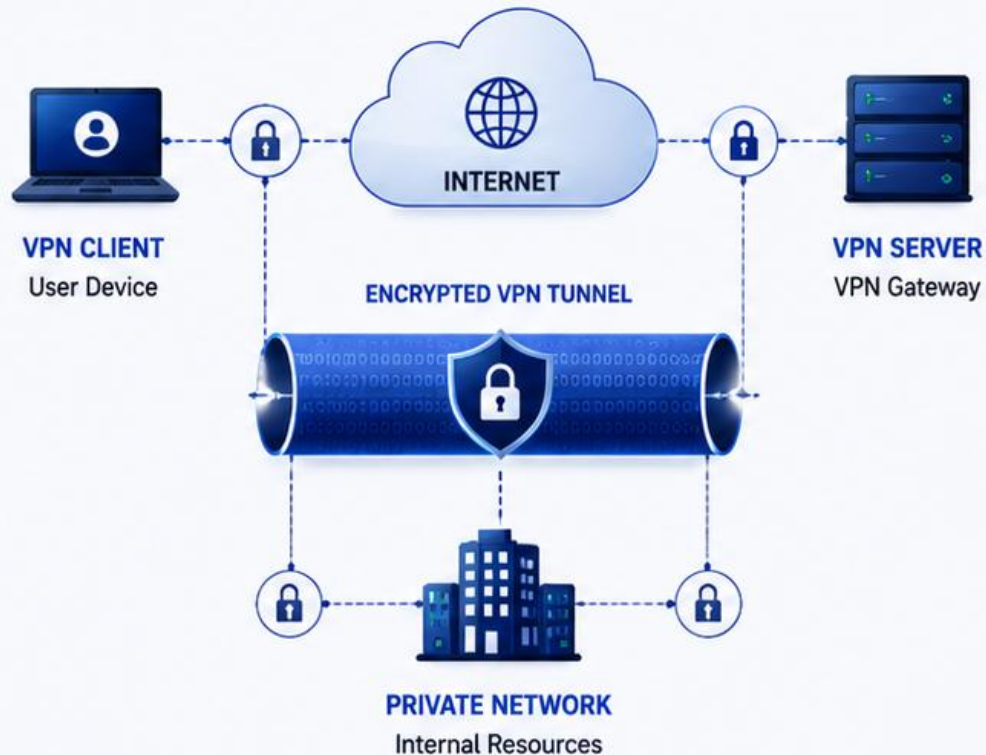
VPN

VIRTUAL PRIVATE NETWORK

VPN COMPONENTS

VPN uses several components and protocols to establish a **secure** and **encrypted** connection over the internet.

VPN ARCHITECTURE OVERVIEW



“

Strong Components.
Secure Connections.
Trusted Networks.



”

KEY VPN COMPONENTS

COMPONENT	DESCRIPTION
 VPN Client	Software or device used by the user to initiate VPN connection.
 VPN Server / Gateway	Server that authenticates users and creates secure VPN tunnels.
 Internet	Public network that carries encrypted data between client and server.
 Encrypted Tunnel	Secure, encrypted connection that protects data from eavesdropping.
 Authentication	Verifies the identity of users before granting access.
 Encryption	Protects data confidentiality using encryption protocols.
 Private Network	Internal network resources accessed securely through VPN.

05

/ 30



VPN

VIRTUAL PRIVATE NETWORK

VPN PORTS & PROTOCOLS

VPN uses specific protocols and ports to establish **secure** and **encrypted** communication over the internet.

COMMON VPN PROTOCOLS & PORTS

PROTOCOL	PORT / PROTOCOL	TRANSPORT	DESCRIPTION
 PPTP	TCP 1723 (Control) GRE (47)	TCP / GRE	Point-to-Point Tunneling Protocol. Uses TCP for control and GRE for data encapsulation.
 L2TP	UDP 1701 (Control) Optional: UDP 500, 4500 (with IPsec)	UDP	Layer 2 Tunneling Protocol. Often used with IPsec for encryption.
 IPsec (ESP)	UDP 50 (ESP)	IP	Encapsulating Security Payload. Provides encryption and data integrity.
 IPsec (AH)	UDP 51 (AH)	IP	Authentication Header. Provides authentication and integrity (no encryption).
 IKE / IKEv1 / IKEv2	UDP 500 UDP 4500 (NAT-T)	UDP	Internet Key Exchange. Used for authentication and secure key exchange.
 OpenVPN	UDP 1194 (Default)	UDP / TCP	Open source VPN protocol. Runs on UDP or TCP.
 SSTP	TCP 443 (HTTPS)	TCP	Secure Socket Tunneling Protocol. Uses HTTPS (TCP 443).
 WireGuard	UDP 51820 (Default)	UDP	Modern, fast, and secure VPN protocol.

“

Right Protocol.
Right Port.
Secure Connection.



”

VPN PROTOCOLS WORK AT DIFFERENT LAYERS



APPLICATION LAYER

- SSTP, OpenVPN
(Work at Application Layer)



TRANSPORT LAYER

- PPTP, L2TP
(Work at Transport Layer)



NETWORK LAYER

- IPsec (AH/ESP)
(Work at Network Layer)



NOTE

Most secure VPNs use IPsec with IKE for authentication and key exchange.

06

/ 30



VPN

VIRTUAL PRIVATE NETWORK

BENEFITS OF VPN

VPN provides **security**, **privacy**, and **flexibility** for individuals and organizations in today's digital world.

“

Privacy is not an option, and it shouldn't be the price we accept for just **getting on the internet.**



”

07

/ 30

KEY BENEFITS



Enhanced Security

Encrypts data and protects from hackers, malware, and cyber threats.



Privacy Protection

Hides your IP address and online activity from ISPs and third parties.



Access Anywhere

Allows secure access to resources and networks from anywhere in the world.



Cost Effective

Reduces costs compared to leased lines and dedicated connections.



Scalability

Easily scalable solution for growing businesses and remote teams.



Data Integrity

Ensures data integrity and prevents eavesdropping or tampering.

BUSINESS BENEFITS



Increased Productivity

Employees can work securely from any location.



Better Collaboration

Securely connect branches, partners, and teams.



Regulatory Compliance

Helps meet data protection and compliance requirements.



Business Continuity

Ensures uninterrupted access to critical systems and data.



Operational Savings

Lower infrastructure costs and simplified network management.

COMMON USE CASES



Remote Work /
Work From Home



Site-to-Site
Connection



E-Learning
Access



Secure Online
Transactions



Safe Public Wi-Fi
Usage



Healthcare
Data Access



VPN

VIRTUAL PRIVATE NETWORK

VPN SECURITY FEATURES

VPN ensures your internet connection is **secure**, **private**, and **protected** from unauthorized access.

HOW VPN PROTECTS YOU



ADVANCED SECURITY FEATURES

FEATURE	DESCRIPTION	BENEFIT
 Strong Encryption	Uses AES-256 bit encryption to secure all data transmissions.	Protects data from hackers and eavesdroppers.
 Authentication	Multi-factor authentication ensures only authorized users can connect.	Prevents unauthorized access.
 No-Logs Policy	We do not store or track your online activities or connection logs.	Ensures complete privacy and anonymity.
 Kill Switch	Automatically disconnects internet if VPN connection drops unexpectedly.	Prevents data leaks and keeps you protected.
 DNS Leak Protection	Prevents DNS requests from being exposed outside the VPN tunnel.	Keeps your browsing activity private.
 Automatic Updates	Keeps VPN software up-to-date with the latest security patches.	Protection against new threats and vulnerabilities.

“

Security is not just a feature,
it's our foundation.

Stay Secure. Stay Connected.



”

08

/ 30

VPN USE CASES



Remote Work

Securely access your company network from anywhere.



Branch Office Connectivity

Connect multiple offices securely over the internet.



Secure Online Transactions

Protect sensitive transactions and customer data.



E-Learning

Access educational resources and platforms securely.



Travel with Security

Stay secure on public Wi-Fi while traveling.



TYPES OF VPN

VPNs can be broadly categorized based on their **access**, **ownership**, and **deployment**.

“

Different needs.
Different VPNs.
One Goal – **Secure Connection**.



”

BASED ON ACCESS

-  **Remote Access VPN**
Allows individual users to connect remotely to a private network.
-  **Site-to-Site VPN**
Connects entire networks or offices to each other securely.

BASED ON OWNERSHIP

-  **Intranet VPN**
Used within an organization to connect different sites of the same organization.
-  **Extranet VPN**
Used to securely connect an organization with external partners or clients.

BASED ON DEPLOYMENT

1. Client-to-Site VPN



2. Site-to-Site VPN



3. Client-to-Client VPN



COMPARISON OVERVIEW

TYPE	USED FOR	EXAMPLE
Remote Access VPN	Individual remote users	Employee working from home
Site-to-Site VPN	Connecting multiple networks	Head office to branch office
Intranet VPN	Internal organization communication	Different departments in same company
Extranet VPN	Secure access for partners/clients	Company and vendor network
Client-to-Site VPN	User connects to company network	Salesperson accessing CRM
Client-to-Client VPN	Direct connection between users	Two employees sharing files



Note: Choosing the right type of VPN depends on your organization's needs, security policies, and network structure.



VPN

VIRTUAL PRIVATE NETWORK



VPN SETUP PROCESS

Setting up a VPN involves several steps to ensure a **secure** and **reliable connection**.

“

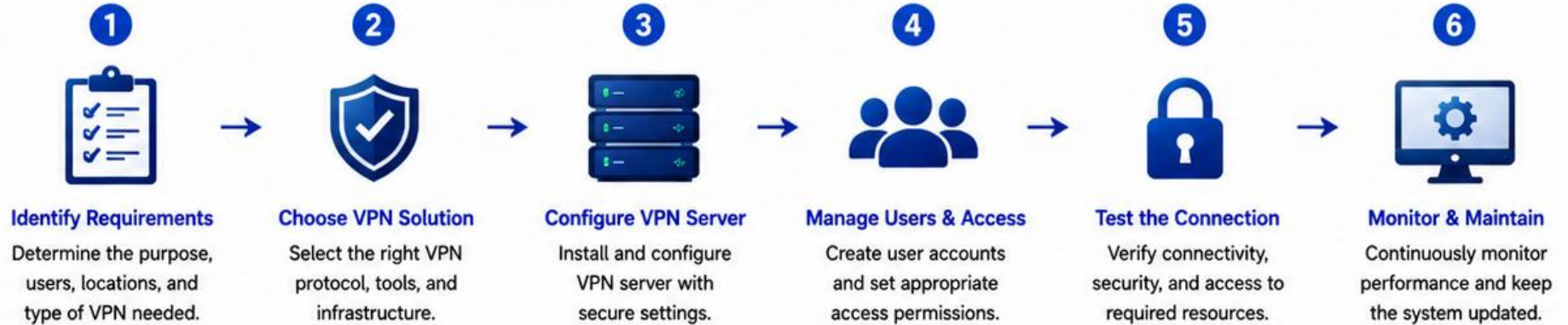
A well-configured VPN means a secure tomorrow.
Setup it right. Stay protected.



”

10
/ 30

STEP-BY-STEP VPN SETUP PROCESS



REQUIREMENTS FOR VPN SETUP

REQUIREMENT	DESCRIPTION
 Reliable Internet Connection	Stable internet is essential for smooth VPN connectivity.
 VPN Server / Gateway	A properly configured server to handle VPN connections.
 User Authentication	Valid credentials or certificates for secure access.
 Firewall & Security Policies	Allow VPN traffic and block unauthorized access.
 Client Software / Device	VPN client software or compatible device for connection.

BEST PRACTICES

- ✓ Use strong authentication methods (MFA, certificates).
- ✓ Keep VPN software and firmware up to date.
- ✓ Use encryption protocols like IPsec or OpenVPN.
- ✓ Implement least privilege access for users.
- ✓ Regularly monitor logs and connection activity.
- ✓ Educate users about VPN security and safe usage.



@manishburdak45



VPN

VIRTUAL PRIVATE NETWORK



QUICK TIP

Always connect to the VPN before accessing sensitive resources.



@manishburdak45

VPN BEST PRACTICES

Following best practices ensures maximum **security**, **performance**, and **reliability** of your VPN connection.



DO'S



Use strong authentication methods (MFA, certificates).



Keep VPN software and firmware up to date.



Use strong encryption protocols (IKEv2, OpenVPN).



Implement least privilege access for users.



Regularly review and update access policies.



Monitor logs and connection activity regularly.



DON'TS



Don't use weak passwords or shared accounts.



Don't download VPN software from unknown sources.



Don't use VPN on unsecured public Wi-Fi without caution.



Don't grant unnecessary access to resources.



Don't ignore updates and security alerts.



Don't disable logging without a valid reason.

“

Good security habits today build a safer digital world tomorrow.



”

11

/ 30

COMMON VPN THREATS



Eavesdropping

Attackers may intercept data on unsecured connections.



Phishing Attacks

Fake VPN sites or emails can steal credentials.



Rogue Wi-Fi

Malicious hotspots can capture your VPN traffic.



Server Compromise

Compromised VPN servers can expose sensitive data.



Malware

Malware can bypass or disable VPN connections.

TIPS FOR OPTIMAL VPN PERFORMANCE



Choose a server closer to your location.



Use high-performance VPN servers with good bandwidth.



Enable split tunneling to route only necessary traffic.



Close unnecessary applications to save bandwidth.



Regularly test connection speed and stability.



Remember: A VPN is only as secure as the practices and policies behind it. Stay informed. Stay secure.



VPN

VIRTUAL PRIVATE NETWORK

VPN USE CASES

VPNs serve a **wide range of purposes** across different users and industries.

“

The right VPN use case can solve real problems, **improve productivity**, and ensure security.



”

12
/ 30

COMMON VPN USE CASES



Remote Work

Securely connect to your company network from anywhere in the world.



Business Connectivity

Connect branch offices, partners, and teams securely over the internet.



Secure Data Access

Access internal resources and sensitive data safely and encrypted.



Access Geo-Restricted Content

Bypass geo-blocks and access websites, streaming services, and content.



E-Learning

Access educational platforms and resources securely from anywhere.



Safe Online Banking & Transactions

Protect your financial transactions on public Wi-Fi and unsecured networks.



Gaming

Reduce latency, protect against DDoS attacks, and access regional servers.

INDUSTRY-SPECIFIC USE CASES



Healthcare

Secure patient data and enable remote access for medical staff.



Finance

Ensure secure communications, transactions, and compliance.



Retail & E-commerce

Secure customer data and connect multiple store locations.



Government

Protect sensitive data and enable secure inter-agency communication.



Manufacturing

Connect IoT devices and remote facilities securely.

VPN BENEFITS ACROSS USE CASES



Enhanced Security

Encrypts data and protects against unauthorized access.



Cost Efficiency

Reduces costs by using the internet instead of private leased lines.



Scalability

Easily scales to accommodate more users, locations, and devices.



Business Continuity

Ensures uninterrupted access to critical resources and services.

CHOOSE THE RIGHT VPN SOLUTION

- ✓ Identify your specific needs and use cases.
- ✓ Evaluate security features and compliance requirements.
- ✓ Consider performance, server locations, and reliability.
- ✓ Ensure ease of use and cross-platform compatibility.
- ✓ Choose a trusted VPN provider with good support.



Key Takeaway: Whether you're an individual, a student, or an enterprise, VPNs provide secure, reliable, and flexible solutions tailored to your needs.



DID YOU KNOW?

VPNs are used not only by individuals but also by businesses, organizations, and governments to secure critical data and communications.



@manishburdak45



VPN

VIRTUAL PRIVATE NETWORK

HOW VPN WORKS

A VPN creates a **secure tunnel** between your device and the internet, protecting your data and hiding your online identity.

“

Behind the scenes, a VPN works to keep you **safe and private**.

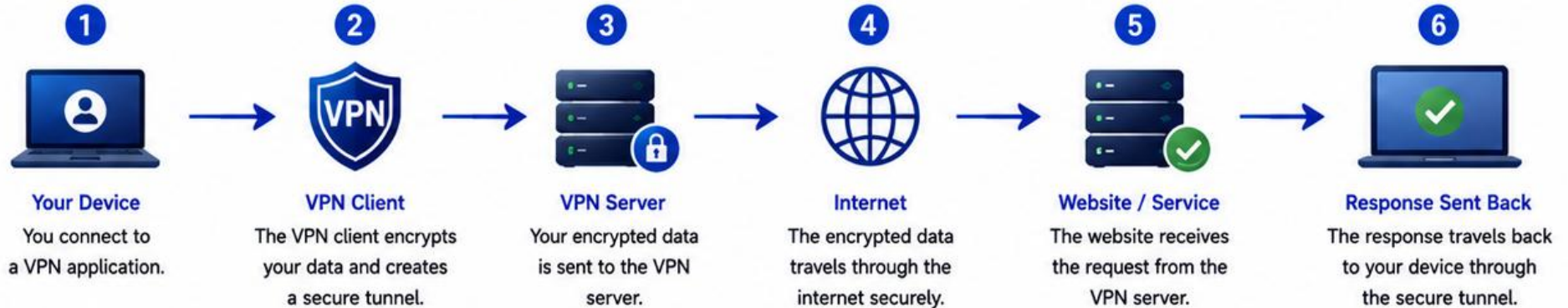


”

13

/ 30

THE VPN CONNECTION PROCESS



WHAT GETS PROTECTED?

-  **Your IP Address** Hidden from websites, apps, and trackers.
-  **Your Online Activity** Encrypted from ISPs, hackers, and third parties.
-  **Your Location** Appears as the VPN server location, not your real one.
-  **Your Data** Protected on public Wi-Fi and untrusted networks.

KEY TECHNOLOGIES INVOLVED

-  **Encryption**
Converts your data into unreadable code to keep it secure.
-  **Tunneling Protocols**
Protocols like OpenVPN, WireGuard, or IKEv2 build the secure tunnel.
-  **Authentication**
Verifies your identity before allowing secure access.
-  **Data Encapsulation**
Your data is wrapped in secure packets and sent through the tunnel.

WHY IT MATTERS

-  **Privacy** Keeps your online activities private and anonymous.
-  **Security** Protects your data from hackers and network snoopers.
-  **Freedom** Access geo-restricted content and websites.
-  **Peace of Mind** Safe browsing anytime, anywhere.



QUICK FACT

A VPN creates a secure tunnel for your data, protecting your privacy and keeping your online activities private.



@manishburdak45



Remember: A VPN acts like a private, encrypted tunnel for your internet connection—keeping your data safe from prying eyes.





KEY TAKEAWAYS

-  VPN protocols are the rules that determine how your data is secured and transmitted.
-  Different protocols offer varying levels of security, speed, and compatibility.
-  Choosing the right protocol depends on your needs, device, and network.

VPN PROTOCOLS EXPLAINED

VPN protocols define how data is **encrypted**, **transmitted**, and **authenticated** between your device and the VPN server.

“The right protocol ensures the perfect balance of **security**, **speed**, and **stability**.”

14
/ 30

 OpenVPN	OpenVPN Open-source and highly secure protocol using SSL/TLS for key exchange and AES encryption.	Best for: Security, reliability, and flexibility.
 IKEv2/IPsec	IKEv2/IPsec Built-in to most devices. Fast, secure, and excellent at reconnecting when network changes.	Best for: Mobile devices and unstable connections.
 WireGuard	WireGuard Modern, lightweight, and faster protocol with strong encryption and small codebase.	Best for: Speed, performance, and modern systems.
 L2TP/IPsec	L2TP/IPsec Uses IPsec for encryption. More secure than L2TP alone but can be slower.	Best for: Compatibility with older devices.
 PPTP	PPTP One of the oldest VPN protocols. Fast but has known security vulnerabilities.	Best for: Streaming and low-security needs only.

VPN PROTOCOL COMPARISON

PROTOCOL	SECURITY	SPEED	STABILITY	COMPATIBILITY	BEST FOR
OpenVPN	★★★★★	★★★☆☆	★★★★☆	★★★★☆	Security & Reliability
IKEv2/IPsec	★★★★☆	★★★★☆	★★★★★	★★★★★	Mobile & On-the-Go
WireGuard	★★★★☆	★★★★★	★★★★☆	★★★★☆	Speed & Performance
L2TP/IPsec	★★★☆☆	★★★☆☆	★★★☆☆	★★★☆☆	Compatibility
PPTP	★☆☆☆☆	★☆☆☆☆	★☆☆☆☆	★★★★★	Streaming & Low Security

HOW TO CHOOSE THE RIGHT PROTOCOL

 Prioritize security? Choose OpenVPN or WireGuard.	 Using mobile or switching networks? Go with IKEv2/IPsec .	 Need the fastest performance? Choose WireGuard .	 Using older devices? Try L2TP/IPsec for compatibility.	 Just streaming or need speed (over security)? PPTP (use with caution).
--	---	--	--	--



Bottom Line: There's no one-size-fits-all protocol. Understand your needs and choose the one that offers the best balance of security, speed, and compatibility for your situation.



@manishburdak45



DID YOU KNOW?

VPNs come in different types, each designed for specific users and scenarios.

@manishburdak45

TYPES OF VPNs

Different types of VPNs are designed to meet **specific needs**.
Choose the one that **fits your requirements best**.

“

Different needs require different VPN solutions.
Pick the right type, get the **right protection**.



”

15
/ 30

VPN TYPE	DESCRIPTION	BEST FOR	KEY FEATURES	EXAMPLE USE CASES
 Remote Access VPN	Allows individual users to connect to a private network from anywhere using the internet.	 Individuals, remote workers, freelancers	✓ Secure remote access ✓ User authentication ✓ Encrypted connection	 Employees accessing company resources from home.
 Site-to-Site VPN	Connects entire networks or multiple locations securely over the internet.	 Businesses with multiple offices or branches	✓ Network-level connection ✓ Always-on connectivity ✓ Cost-effective for scale	 Connecting head office with branch offices securely.
 Mobile VPN	Provides secure access to network resources from mobile devices.	 Mobile users, field employees, travelers	✓ Secure on-the-go access ✓ Works on mobile networks ✓ Lightweight & flexible	 Sales teams accessing CRM while traveling or on the move.
 SSL/TLS VPN	Uses SSL/TLS encryption through a web browser. No special client required.	 Users needing quick and easy access	✓ Browser-based access ✓ Easy to deploy ✓ Cross-platform support	 Accessing internal web applications securely via browser.
 Cloud VPN	Uses cloud-based gateways to provide secure and scalable connectivity.	 Organizations using cloud services	✓ Scalable and flexible ✓ High availability ✓ Cost-efficient	 Securely connecting to cloud infrastructure (AWS, Azure, GCP).



Pro Tip: Evaluate your needs, infrastructure, and budget before choosing a VPN type.
The right choice ensures security, performance, and peace of mind.



VPN RISKS & HOW TO STAY SAFE

While VPNs offer many benefits, they are **not risk-free**. Understanding the risks helps you take the right steps to stay protected.

“

Security is not a product, it's a practice. Stay **aware**, stay safe.



”

16
/ 30

COMMON VPN RISKS



Untrusted VPN Providers

Some VPNs may log your data, inject ads, or even sell your information to third parties.



Data Leaks

IP, DNS, or WebRTC leaks can expose your real identity even while using a VPN.



Weak Encryption

Outdated or weak encryption protocols can be cracked by attackers.



Slow Speeds

VPNs can reduce internet speed due to encryption and server distance.



Server Risks

Compromised VPN servers can expose your data or redirect your traffic.



Legal & Policy Issues

VPN use may be restricted or illegal in certain countries or situations.

HOW TO STAY SAFE

1



Choose a Reputable VPN

Select a trusted provider with a no-logs policy and strong privacy practices.

2



Use Strong Encryption

Ensure your VPN uses AES-256 encryption and secure protocols like OpenVPN or WireGuard.

3



Check for Leaks Regularly

Use leak testing tools to ensure your IP, DNS, and WebRTC information stay private.

4



Use Kill Switch

Enable the kill switch feature to block internet traffic if the VPN connection drops.

5



Keep Software Updated

Regularly update your VPN app and device OS to patch security vulnerabilities.

6



Connect to the Right Server

Choose servers that are closer to your location for better speed and reliability.

7



Practice Safe Online Habits

Avoid suspicious links, use strong passwords, and enable two-factor authentication.



QUICK TIP

A VPN adds a layer of security, but safe online behavior is still in your hands.

VPN SAFETY CHECKLIST



VPN provider has a clear no-logs policy.



Strong encryption and secure protocols are used.



No IP, DNS, or WebRTC leaks detected.



Kill switch is enabled.



VPN app and device are up to date.



Connected to a trusted VPN server.



Remember: A VPN enhances your privacy and security, but staying safe online is a combination of the right tools, smart choices, and constant awareness.



@manishburdak45

HOW TO SET UP A VPN

Setting up a VPN is simple. Follow these steps to get started and **secure your connection** in minutes.

“

A few simple steps today can **protect your privacy** for years to come.



”

17
/ 30

7 EASY STEPS

1



Choose a VPN

Select a reputable VPN provider that meets your needs and budget.

2



Download the App

Go to the official website and download the VPN app for your device.

3



Install the App

Install the VPN application and grant necessary permissions.

4



Sign In / Create an Account

Log in to your account or create a new one.

5



Choose a Server

Select a server location based on your needs (speed, privacy, or access).

6



Connect

Click the connect button and let the VPN secure your connection.

7



You're Protected!

Your traffic is now encrypted and your online activity is private.

SET UP ON POPULAR DEVICES



Windows

Download the app, install it, log in, and connect to a server.



macOS

Drag the app to Applications, open, log in, and connect.



Android

Install from Google Play Store, open, log in, and connect.



iOS

Install from App Store, open, log in, and connect.



Linux

Install using your distro's package manager or setup file.



Router

Access router settings, configure VPN, and connect all devices.



DID YOU KNOW?

Setting up a VPN is usually quick and easy. The right setup ensures maximum security and the best experience.

BEST PRACTICES

- ✓ Always use trusted VPN providers.
- ✓ Enable auto-connect on trusted Wi-Fi.
- ✓ Keep your VPN app updated.
- ✓ Use strong passwords and 2FA.
- ✓ Disconnect when VPN is not needed (to save battery and bandwidth).



BENEFITS YOU GET



Online Privacy

Hide your IP address and stay anonymous.



Secure Data

Encrypt your data and protect it from hackers.



Access Worldwide Content

Unblock websites and streaming services.

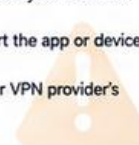


Safer Connections

Stay protected on public Wi-Fi and networks.

TROUBLESHOOTING TIPS

- ⚠ Can't connect? Try another server.
- ⚠ Slow speed? Choose a closer server.
- ⚠ Connection drops? Check your internet.
- ⚠ Still not working? Restart the app or device.
- ⚠ Need help? Contact your VPN provider's support team.



Final Tip: A VPN is a powerful tool for privacy and security. Set it up once, use it wisely, and enjoy a safer internet every day.



VPN USE CASES

VPNs serve a **wide range of needs**. Here are the most common ways people and organizations use VPNs.

“

A VPN adapts to your life—protecting what matters most, **wherever you are**.



”

1 ONLINE PRIVACY & ANONYMITY



Hide your IP address and encrypt your connection to keep your online activities private.

Best for:

Everyday users who value privacy.

2 SECURE REMOTE WORK



Access company resources securely from anywhere without risking data exposure.

Best for:

Remote employees and businesses.

3 ACCESS RESTRICTED CONTENT



Bypass geo-restrictions and access websites, streaming platforms, and content globally.

Best for:

Streamers, travelers, and content enthusiasts.

4 SAFER PUBLIC WI-FI



Encrypt your data on public networks to prevent hackers from intercepting it.

Best for:

Travelers, students, and coffee shop workers.

5 GAMING ADVANTAGE



Reduce lag, avoid DDoS attacks, and connect to gaming servers in different regions.

Best for:

Gamers and eSports players.

6 BUSINESS & ENTERPRISE USE



Connect multiple offices, protect communications, and ensure secure data transfers.

Best for:

Enterprises and organizations.



DID YOU KNOW?

VPNs are used by individuals, businesses, and organizations worldwide to protect data, access content, and work securely from anywhere.



KEY TAKEAWAY

A VPN gives you privacy, security, and freedom online.



@manishburdak45

WHY VPNs ARE ESSENTIAL TODAY



Protects Your Privacy

Keeps your personal information and browsing habits private.



Secures Your Data

Encrypts your connection to protect sensitive data.



Gives You Freedom

Access the internet without borders or restrictions.



Saves Money

Find better deals and prices by changing your virtual location.



Enhances Security

Adds an extra layer of protection against cyber threats.

VPN vs NO VPN

WITH VPN

- ✓ Your IP address is hidden
- ✓ Your data is encrypted
- ✓ Your online activity is private
- ✓ Access to global content
- ✓ Safer on public Wi-Fi

VS

WITHOUT VPN

- ✗ Your IP address is visible
- ✗ Your data is not encrypted
- ✗ Your activity can be tracked
- ✗ Limited content access
- ✗ Risky on public Wi-Fi



Pro Tip: No matter your goal—privacy, security, access, or performance—a VPN is a smart and simple solution that puts you back in control of your online experience.





DID YOU KNOW?

VPNs were originally designed for businesses to secure communication over private networks, but now they're used by millions worldwide every day.

KEY TAKEAWAY

A VPN protects your data, unlocks content, and gives you more freedom online.

@manishburdak45

VPN FEATURES EXPLAINED

Modern VPNs come with powerful features that enhance your security, privacy, and overall experience.

“The right features make a good VPN great. Know what you're getting.”

- 1 

Strong Encryption

Encrypts your data with AES-256 or higher to keep it safe from hackers and snoopers.
- 2 

Global Server Network

Access servers in multiple countries to unlock content and get better speeds.
- 3 

No-Logs Policy

Ensures your online activity is not recorded or stored by the VPN provider.
- 4 

High-Speed Connection

Optimized servers and protocols deliver fast and stable connections for streaming and more.
- 5 

Multi-Device Support

Protect multiple devices with one account—phones, laptops, tablets, and more.
- 6 

Kill Switch

Automatically blocks internet if the VPN connection drops to prevent data leaks.
- 7 

Split Tunneling

Choose which apps use the VPN and which access the internet directly.
- 8 

DNS Leak Protection

Prevents DNS requests from leaking outside the VPN tunnel and exposing your activity.
- 9 

Auto-Connect

Automatically connects you to the VPN when you join untrusted networks.
- 10 

Ad & Tracker Blocker

Blocks ads, trackers, and malware to give you a cleaner and safer browsing.

EXTRA FEATURES TO LOOK FOR

-  **Dedicated IP**

Get a unique IP address for secure remote access and fewer blocks.
-  **Obfuscated Servers**

Hide VPN usage and bypass VPN blocks in restricted networks.
-  **Streaming Support**

Optimized servers to unblock Netflix, Hulu, Disney+, and more.
-  **Malware Protection**

Block malicious websites and downloads before they reach you.
-  **24/7 Customer Support**

Get help anytime via live chat, email, or support tickets.

POPULAR VPN PROTOCOLS

PROTOCOL	SPEED	SECURITY	BEST FOR	NOTES
OpenVPN (UDP)	★★★★☆	★★★★★	All-around use	Open-source, highly secure and reliable.
WireGuard	★★★★★	★★★★☆	Speed & performance	Modern protocol, lightweight and fast.
IKEv2/IPsec	★★★★☆	★★★★☆	Mobile devices	Very stable, great for switching networks.
L2TP/IPsec	★★★☆☆	★★★☆☆	Compatibility	Slower but works on older devices.
PPTP	★☆☆☆☆	★☆☆☆☆	Legacy use only	Outdated and not secure. Avoid if possible.



Remember: Features matter, but privacy and security should always come first. Choose a VPN that aligns with your needs and offers the features that make the biggest difference to you.





HOW VPNs WORK

A VPN creates a secure tunnel between your device and a VPN server, keeping your data **private** and **protected**.

“

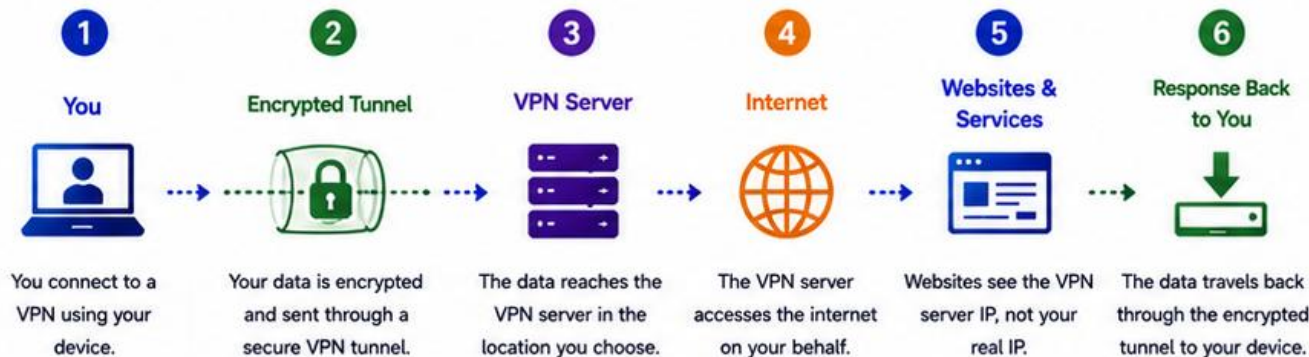
Behind the scenes, VPNs work to keep your connection **secure**, **private**, and **anonymous**.



”

20
/ 30

THE VPN CONNECTION PROCESS



Result: Your IP address is hidden, your data is encrypted, and your online activity stays private.

VPN vs REGULAR CONNECTION

REGULAR CONNECTION (NO VPN)



- ✗ Your ISP can see your activity
- ✗ Your real IP address is visible
- ✗ Your data is not encrypted
- ✗ You are easier to track and profile

VPN CONNECTION



- ✓ Your activity is private from your ISP
- ✓ Your data is encrypted
- ✓ Your real IP is hidden
- ✓ You are harder to track

TIPS TO GET THE MOST OUT OF YOUR VPN



Choose the Right Server

Pick a server close to you for speed, or in another country for access.



Use Fast Protocols

WireGuard and IKEv2 offer the best speed and security.



Keep It On

Keep your VPN connected all the time for continuous protection.



Update Regularly

Keep your VPN app updated for the latest features and security fixes.



Check for Leaks

Use leak test tools to ensure your IP, DNS, and WebRTC are secure.



Read the Privacy Policy

Choose VPNs with clear no-logs policies you can trust.



DID YOU KNOW?

Without a VPN, your ISP, websites, and others can see your online activity. A VPN hides it from prying eyes.



KEY TAKEAWAY

A VPN acts like a secure tunnel for your data, keeping it private from start to finish.



@manishburdak45



Bottom Line: A VPN secures your connection, hides your identity, and gives you freedom online. It's like a private tunnel between you and the internet.



CISCO IOS INTRODUCTION

Cisco IOS (Internetwork Operating System) is the software that runs on Cisco routers and switches. It provides the foundation for **network connectivity, management, and security**.

“Cisco IOS powers **secure, intelligent, and resilient** networks worldwide.”



WHAT IS CISCO IOS?



- ✓ Cisco IOS is the operating system that runs on Cisco networking devices.
- ✓ It handles routing, switching, security, and device management.
- ✓ Provides a stable, secure, and scalable platform for enterprise networks.
- ✓ Continuously updated with new features, security fixes, and performance improvements.

ROUTER & SWITCH MANAGEMENT



ROUTER

- ✓ Routing protocols
- ✓ NAT & ACL
- ✓ VPN & Security
- ✓ Interface configuration
- ✓ System monitoring



SWITCH

- ✓ VLANs & Trunking
- ✓ STP & EtherChannel
- ✓ Port security
- ✓ QoS & Traffic control
- ✓ System monitoring

CLI vs GUI

CLI (Command Line Interface)



- ✓ Powerful and flexible
- ✓ Faster configuration
- ✓ Scriptable & automatable
- ✓ Preferred by professionals

GUI (Graphical User Interface)



- ✓ User-friendly
- ✓ Visual monitoring
- ✓ Easy for beginners
- ✓ Limited advanced control

VS

ROLE OF IOS IN NETWORK SECURITY

- Access Control**
ACLs, AAA, and RBAC restrict unauthorized access.
- Data Protection**
Encryption, VPNs, and secure protocols protect data in transit.
- Threat Prevention**
Features like IP Source Guard, Port Security, and uRPF prevent attacks.
- Monitoring & Logging**
Syslog, SNMP, and NetFlow help detect and respond to threats.
- Secure Management**
SSH, HTTPS, and SNMPv3 ensure safe device management.

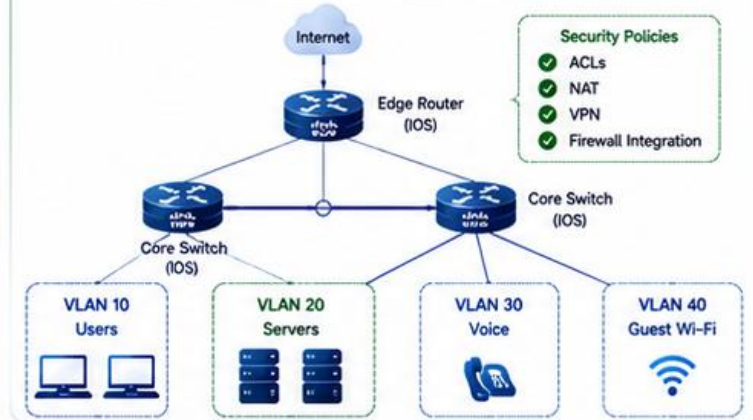
COMMON CISCO DEVICES USING IOS



IOS MODES (CLI HIERARCHY)



ENTERPRISE NETWORK EXAMPLE



DID YOU KNOW?

Cisco IOS powers millions of network devices worldwide and is one of the most trusted operating systems in networking.

★ KEY TAKEAWAY

Cisco IOS is the foundation that makes Cisco devices secure, reliable, and scalable.



In Short: Cisco IOS is the brain of Cisco networking devices. It enables connectivity, optimizes performance, and strengthens security—making it the backbone of modern enterprise networks.



CISCO IOS INTRODUCTION

Cisco IOS (Internetwork Operating System) is the heart of Cisco networking devices. It enables **connectivity**, **management**, and **security** for reliable and efficient networks.

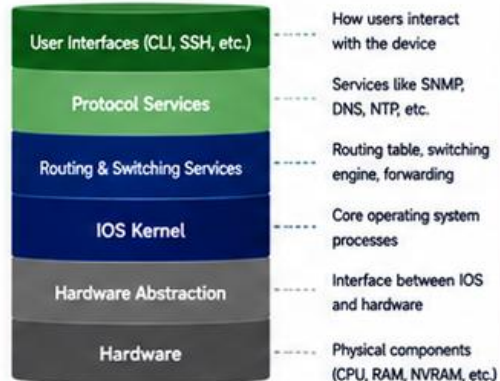
“Powerful software. Reliable networks. Secure connections. That’s the power of **Cisco IOS**.”



KEY CAPABILITIES OF IOS

- Routing & Switching**
Supports dynamic routing protocols and advanced switching features.
- Security**
Provides AAA, ACLs, VPNs, firewall, and more to protect networks.
- Management**
Offers CLI access, SNMP, logging, monitoring, and automation.
- Scalability & Reliability**
Designed to grow with your network and deliver high availability.

IOS ARCHITECTURE (HIGH LEVEL)



CLI vs GUI (MANAGEMENT)

CLI (Command Line)



- ✓ Text-based interface
- ✓ Fast and resource-efficient
- ✓ Full control & automation
- ✓ Ideal for advanced users and scripting

VS

GUI (Web-based)



- ✓ Visual interface
- ✓ Easy to use
- ✓ Good for monitoring and basic tasks
- ✓ Ideal for beginners and casual use

COMMON IOS MODES (CLI)

- User EXEC Mode**: Basic monitoring (>)
- Privileged EXEC Mode**: Full device access (#)
- Global Configuration Mode**: Configure the entire device (conf)
- Interface Configuration Mode**: Configure interface settings (intf)#
- Line Configuration Mode**: Configure console, vty, ring-line)#

DID YOU KNOW?

Cisco IOS was introduced in the late 1980s and has evolved for 30+ years, powering billions of devices and networks worldwide.

★ KEY TAKEAWAY

Cisco IOS gives you the power to configure, manage, and secure networks with precision and control.

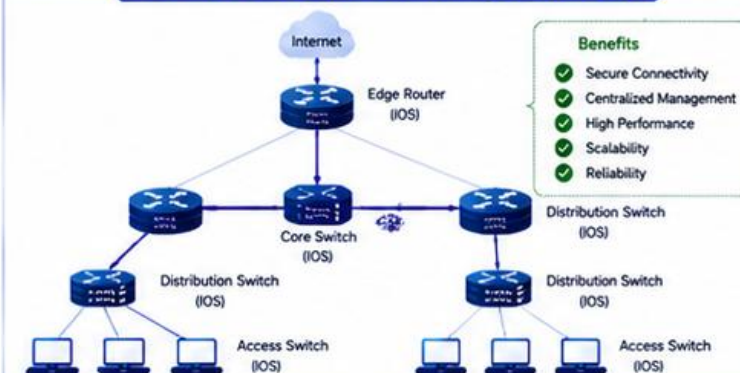
IOS FILE SYSTEM (IMPORTANT FILES)

FILE	DESCRIPTION
 startup-config	Saved configuration that loads on boot.
 running-config	Active configuration in RAM.
 IOS Image	The operating system file.
 flash:	Internal storage for images and files.
 NVRAM	Non-volatile memory for configs.

COMMON IOS COMMANDS (EXAMPLES)

COMMAND	PURPOSE
show ip interface brief	View IP interfaces and status.
show running-config	Display current running configuration.
show ip route	Show routing table.
ping <ip-address>	Test connectivity to a host.
copy running-config startup-config	Save configuration to NVRAM.

TYPICAL ENTERPRISE NETWORK (IOS DEVICES)





CISCO IOS CLI BASICS

CISCO IOS CLI BASICS

The Cisco IOS Command Line Interface (CLI) allows you to **configure**, **manage**, and **troubleshoot** Cisco routers and switches efficiently.

“

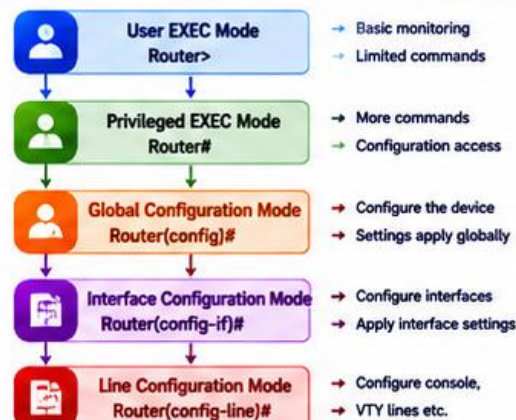
Master the CLI—
your most **powerful** tool
for **network control**.



”

23
/ 30

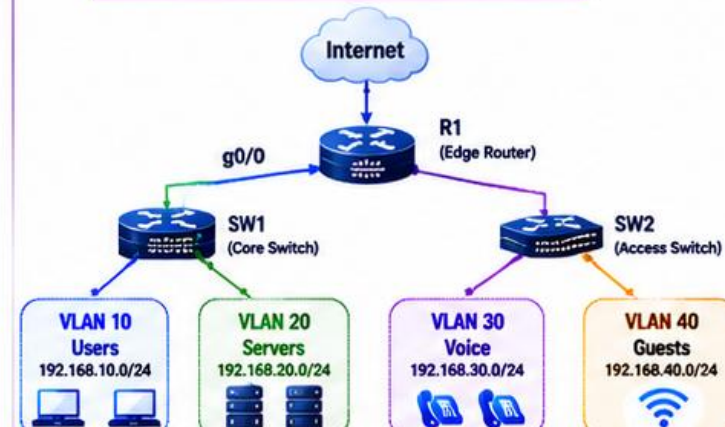
CISCO CLI MODES HIERARCHY



BASIC IOS COMMANDS (EXAMPLES)

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config-if)# interface g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.254
R1(config)# end
R1# write memory
```

TYPICAL NETWORK TO PRACTICE



SHOW COMMANDS (MONITORING)

```
R1# show ip interface brief
R1# show ip route
R1# show running-config
R1# show vlan brief
R1# show interfaces g0/0
R1# show cdp neighbors
R1# show version
```

INTERFACE CONFIGURATION

```
R1(config)# interface g0/0
R1(config-if)# description Uplink_to_SW1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

VLAN CONFIGURATION (SWITCH)

```
SW1(config)# vlan 10
SW1(config-vlan)# name Users
SW1(config)# vlan 20
SW1(config-vlan)# name Servers
SW1(config)# interface range fa0/1 - 10
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
```

SECURITY BEST PRACTICES



SAVE & CONFIG MANAGEMENT



TROUBLESHOOTING TIPS

- Check connectivity → ping, tracer
- Check routes → show ip route
- Check interfaces → show ip int brief
- Check VLANs → show vlan brief
- Check neighbors → show cdp neighbors



DID YOU KNOW?

Cisco IOS CLI is one of the most powerful tools for configuring and managing network devices with precision.



KEY TAKEAWAY

- Learn the CLI modes.
- Use basic IOS commands.
- Configure interfaces, VLANs.
- Secure and save configs.
- Troubleshoot with show commands.



@manishburdak45



Pro Tip: Practice CLI daily. The more you type, the faster and smarter you become. The CLI gives you **complete control** over your Cisco devices.





DID YOU KNOW?

VLANs help segment a network logically, improving security, performance, and management without needing extra hardware.

KEY TAKEAWAY

- ✓ VLANs isolate broadcast domains.
- ✓ They improve security.
- ✓ They optimize performance.
- ✓ They simplify network management.

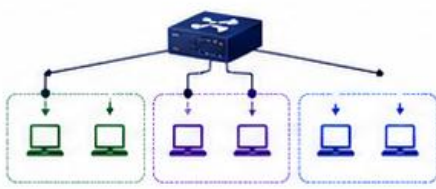
VLAN BASICS & SECURITY

A VLAN (Virtual Local Area Network) logically segments a network into multiple broadcast domains. It improves **security**, **performance**, and **manageability**.

WHAT IS A VLAN?

A VLAN is a logical network segment that groups devices together, regardless of their physical location.

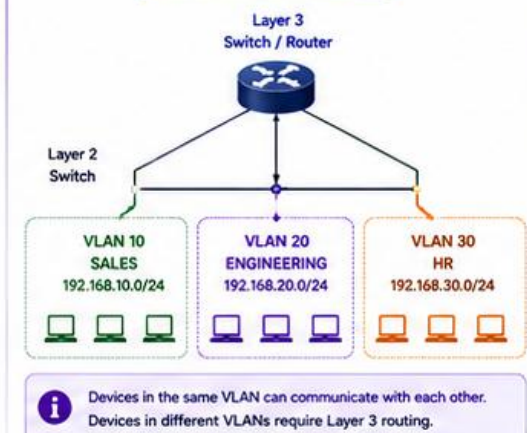
- ✓ Creates separate broadcast domains
- ✓ Reduces broadcast traffic
- ✓ Enhances security and performance



TYPES OF VLANs

VLAN TYPE	DESCRIPTION
Data VLAN	Carries user data. Example: VLAN 10, 20, 30
Default VLAN	VLAN 1 by default. Used for switch management.
Management VLAN	Used for managing network devices securely.
Native VLAN	Carries untagged traffic on trunk links.
Voice VLAN	Used for IP phones and voice traffic.

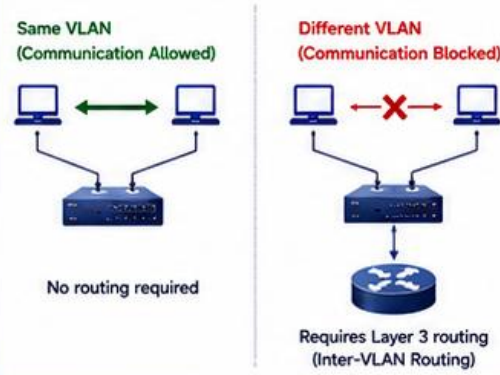
VLAN OPERATION



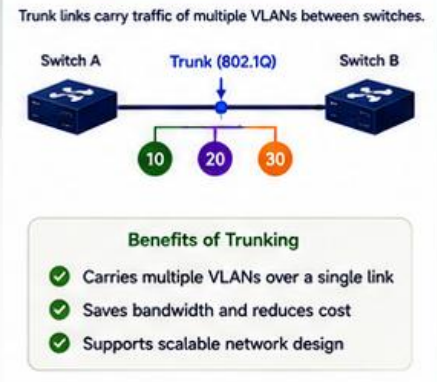
VLAN BENEFITS

- Improved Security
Isolates departments and limits unauthorized access.
- Better Performance
Reduces broadcast traffic and network congestion.
- Easier Management
Logical grouping simplifies network administration.
- Cost Effective
Achieves segmentation without extra physical infrastructure.
- Scalability
Easily add, move or change users without major reconfiguration.

VLAN COMMUNICATION (OVERVIEW)



VLAN TRUNKING (802.1Q)



VLAN SECURITY BEST PRACTICES

- Disable Unused VLANs
Remove unnecessary VLANs to reduce attack surface.
- Restrict VLAN Access
Limit access using ACLs and port security.
- Use Private VLANs (PVLAN)
For advanced isolation within the same VLAN.
- Secure Management VLAN
Use dedicated VLAN and strong authentication.
- Monitor VLAN Traffic
Regularly monitor and log for suspicious activity.
- Keep Firmware Updated
Keep switches and routers updated for security patches.

COMMON VLAN COMMANDS (IOS)

```
! Create VLAN
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config)# vlan 20
Switch(config-vlan)# name ENGINEERING
Switch(config)# vlan 30
Switch(config-vlan)# name HR

! Assign access port
Switch(config)# interface range fa0/1 - 12
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10

! Trunk port
Switch(config)# interface gi0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

Bottom Line: VLANs help you build **secure**, **efficient**, and **scalable** networks. Segment smart, manage easy, secure stronger!





CISCO IOS & VLAN SECURITY

VLAN SECURITY

VLAN SECURITY & BEST PRACTICES

Securing VLANs ensures that data is **protected**, access is **controlled**, and unauthorized users cannot exploit network **segmentation**.

“

A **secure** VLAN is a **secure** network. Protect your **segments**, protect your **data**.



”

25
/ 30

VLAN SECURITY THREATS



VLAN Hopping

Attackers move between VLANs to access restricted resources.



Unauthorized Access

Users gain access to VLANs they should not be in.



Broadcast Storms

Excessive broadcasts consume bandwidth and affect performance.



Misconfiguration

Incorrect configurations can lead to security gaps.

VLAN SECURITY FEATURES



Port Security

Limits the number of MAC addresses on a switch port.



DHCP Snooping

Prevents rogue DHCP servers from assigning IP addresses.



Dynamic ARP Inspection (DAI)

Prevents ARP spoofing and man-in-the-middle attacks.



Private VLAN (PVLAN)

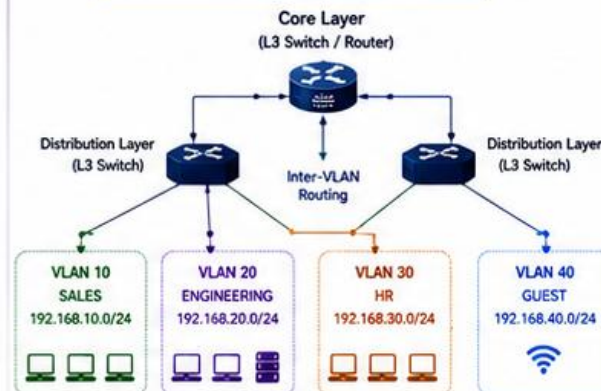
Provides Layer 2 isolation within VLANs.



Access Control Lists (ACL)

Filters traffic between VLANs at Layer 3.

VLAN SECURITY ARCHITECTURE



BEST PRACTICES



Use Strong Access Control
Apply ACLs and port security to restrict unauthorized access.



Enable DHCP Snooping
Allows only trusted DHCP servers to assign IP addresses.



Enable DAI
Protects against ARP spoofing and injection attacks.



Use PVLANs
Isolate users within the same VLAN for better security.



Disable Unused Ports
Shut down unused switch ports to prevent attacks.



Monitor & Log
Regularly monitor VLAN traffic and review logs for threats.



DID YOU KNOW?

VLANs reduce the size of broadcast domains, which improves performance and adds an extra layer of security by isolating traffic.



KEY TAKEAWAY

- ✓ Use VLANs to segment networks.
- ✓ Apply security features to protect VLANs.
- ✓ Control traffic between VLANs.
- ✓ Always follow security best practices.

VLAN CREATION EXAMPLE (IOS CLI)

```
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name ENGINEERING
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# end
Switch# show vlan brief
```

VLAN	Name	Status	Ports
10	SALES	active	Fa0/1, Fa0/2
20	ENGINEERING	active	Fa0/3, Fa0/4
30	HR	active	Fa0/5, Fa0/6

PORT SECURITY CONFIGURATION

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security violation restrict
Switch(config-if)# exit
```

```
Switch# show port-security interface fa0/1

Port Security      : Enabled
Port Status        : Secure-up
Violation Mode      : Restrict
Maximum MAC Addresses : 2
Total MAC Addresses : 2
```

ACL BETWEEN VLANs (EXAMPLE)

```
ip access-list extended BLOCK_HR_SALES
deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
permit ip any any

!
interface vlan 30
ip access-group BLOCK_HR_SALES in
!
interface vlan 10
ip access-group BLOCK_HR_SALES out
```

Explanation:
This ACL blocks HR (VLAN 30) from accessing SALES (VLAN 10) but allows other traffic.

DHCP SNOOPING & DAI (EXAMPLE)

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 10,20,30
Switch(config)# ip dhcp snooping trust interface gi0/1
!
Switch(config)# ip arp inspection vlan 10,20,30
Switch(config)# interface range fa0/1 - 24
Switch(config-if-range)# ip dhcp snooping limit rate 15
Switch(config-if-range)# ip arp inspection limit rate 15
Switch(config-if-range)# end
```



Benefit:
Prevents rogue DHCP servers and ARP attacks, keeping your VLANs secure.



@manishburdak45



Bottom Line: Secure your VLANs, secure your network. Apply **segmentation**, access control, and monitoring to build a **strong** and reliable network infrastructure.





DID YOU KNOW?

VLANs allow you to group devices logically, not physically. This improves security, performance, and simplifies network management.

KEY TAKEAWAY

- ✓ VLANs = Logical Segmentation
- ✓ Improve Security & Performance
- ✓ Use ACLs & Port Security
- ✓ Monitor & Manage Continuously
- ✓ Follow Best Practices

VLAN FUNDAMENTALS

VLANs (Virtual Local Area Networks) allow you to **logically group** devices together in the **same broadcast domain**, regardless of their **physical location**.

VLANs are the foundation of a **secure, efficient, and scalable** enterprise network.

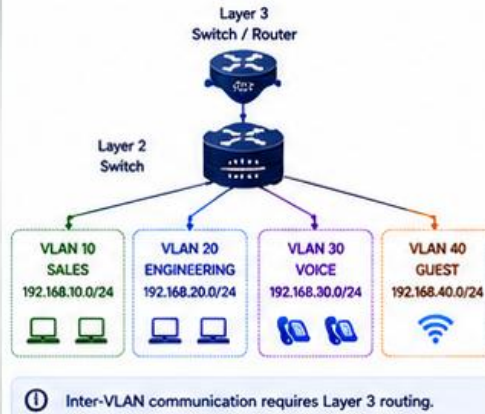
WHAT IS A VLAN?

A VLAN is a logical network segmentation that creates separate broadcast domains within the same physical network.

- ✓ Reduces broadcast traffic
- ✓ Improves security
- ✓ Better performance
- ✓ Simplifies management
- ✓ Cost-effective segmentation



VLAN OPERATION OVERVIEW



VLAN CHARACTERISTICS

- Broadcast Domain**
Each VLAN is a separate broadcast domain.
- Isolation**
Devices in different VLANs cannot communicate at Layer 2.
- Flexibility**
Devices can belong to different VLANs based on role, department, or function.
- Scalability**
Easy to add, move, or change VLANs without physical reconfiguration.

VLAN TYPES

VLAN TYPE	DESCRIPTION
Default VLAN (VLAN 1)	All ports are members by default. Used for management.
Data VLAN	Carries user data (e.g., VLAN 10, 20, 30).
Voice VLAN	Carries voice traffic (optimized for IP phones).
Management VLAN	Used for device management and administration.
Native VLAN	Carries untagged traffic on 802.1Q trunk links.

CREATE VLAN (IOS CL) EXAMPLE

```
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config)# vlan 20
Switch(config-vlan)# name ENGINEERING
Switch(config)# vlan 30
Switch(config-vlan)# name VOICE
Switch(config)# vlan 40
Switch(config-vlan)# name GUEST
Switch(config)# end
Switch# show vlan brief
```

VLAN	Name	Status	Ports
---	default	active	Fa0/1, Fa0/2, Fa0/3
10	SALES	active	Fa0/4, Fa0/5
20	ENGINEERING	active	Fa0/6, Fa0/7
30	VOICE	active	Fa0/8
40	GUEST	active	Fa0/9

ASSIGN PORTS TO VLAN

```
Switch# configure terminal
! Access port (untagged)
Switch(config)# interface fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
!
Switch(config)# interface fa0/6
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
!
Switch(config)# interface fa0/8
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config)# end
```

TRUNK PORT CONFIGURATION (802.1Q)

```
Switch# configure terminal
Switch(config)# interface fa0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk
encapsulation dot1q
Switch(config-if)# switchport trunk allowed vlan
10,20,30,40
Switch(config-if)# switchport trunk native vlan 1
Switch(config)# end
Switch# show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/24	10,20,30,40

ACCESS VS TRUNK PORTS

ACCESS PORT	TRUNK PORT
✓ Connects end devices (PC, IP Phone, Printer) ✓ Carries traffic for one VLAN only ✓ Untagged traffic ✓ Not used to connect switch-to-switch (or router)	✓ Connects network devices (switch, router, firewall) ✓ Carries traffic for multiple VLANs ✓ Tagged traffic (802.1Q) ✓ Used for inter-VLAN communication



Bottom Line: VLANs help you segment networks logically, improve security, reduce broadcast traffic, and make management easier. Proper design and configuration create a strong, **scalable network!**







DID YOU KNOW?

Port Security helps prevent unauthorized devices from accessing your network by limiting the number of MAC addresses on a switch port.

- 

KEY TAKEAWAY

 - Limit MAC addresses per port.
 - Prevent unauthorized access.
 - Choose violation action wisely.
 - Monitor and maintain regularly.

PORT SECURITY BASICS

Port Security is a Layer 2 security feature that **restricts** the number of MAC addresses allowed on a switch port. It helps prevent **unauthorized** devices from connecting.

“

Secure your ports,
secure your network.

Port Security is your **first** line of defense.

”



27

/ 30

WHAT IS PORT SECURITY?

Port Security limits the number of MAC addresses that can be learned on a switch port.

- Works at Layer 2 (Data Link Layer)
- Protects against MAC flooding attacks
- Allows only authorized devices
- Easy to configure and manage



Ideal for access ports (not for trunk ports).

HOW IT WORKS

Switch Port with Port Security

If an unknown MAC address tries to connect, the switch takes action based on the violation mode.

PORT SECURITY MODES

MODE	DESCRIPTION
 protect	Drop packets from unauthorized MAC addresses silently.
 restrict	Drop packets and log the violation (SNMP trap).
 shutdown	Shutdown the port when violation occurs.
 none (default)	No security. Unlimited MAC addresses.

PORT SECURITY BENEFITS

-  **Prevents Unauthorized Access**
Blocks unknown devices from accessing the network.
-  **Reduces Attack Surface**
Helps stop MAC flooding and other Layer 2 attacks.
-  **Improves Network Control**
Allows only trusted devices on the network.
-  **Easy to Implement**
Simple configuration with powerful security impact.

1. BASIC PORT SECURITY CONFIGURATION

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security violation restrict
Switch(config-if)# end
```



Allows max 2 MAC addresses, learns them dynamically, and sets violation mode to restrict.

2. VERIFY PORT SECURITY

```
Switch# show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Restrict
Maximum MAC Addresses  : 2
Total MAC Addresses    : 2
Configured MAC Addresses : 0
Sticky MAC Addresses   : 2
```

```
Switch# show port-security interface fa0/1 mac-address
Vlan  Mac Address      Type      Ports
----  -
10    0011.2233.4455    Sticky   Fa0/1
10    00aa.bb00.ccee    Sticky   Fa0/1
```

3. MANUAL (STATIC) MAC CONFIGURATION

```
Switch(config)# interface fa0/2
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 1
Switch(config-if)# switchport port-security mac-address 00aa.bb00.ccee
Switch(config-if)# switchport port-security violation shutdown
Switch(config-if)# end
```



Allows only the specified MAC address on the port. Any other MAC will cause shutdown.

4. CLEAR PORT SECURITY

```
Switch(config)# interface fa0/1
Switch(config-if)# no switchport port-security
Switch(config-if)# end

Or

Switch# clear port-security interface fa0/1
```



Use this to clear sticky MAC addresses or remove port-security.

PORT SECURITY BEST PRACTICES

- Enable on access ports only.
- Use 'sticky' MAC learning for dynamic devices.
- Set appropriate violation mode (restrict or shutdown).
- Regularly monitor and clear old MAC addresses.
- Document authorized devices.

COMMON USE CASES



QUICK REFERENCE COMMANDS

switchport port-security	Enable Port Security
switchport port-security maximum <n>	Set max MAC addresses
switchport port-security mac-address sticky	Enable sticky learning
switchport port-security mac-address <mac>	Add static MAC address
switchport port-security violation {protect restrict shutdown}	
show port-security interface <int>	Show port-security status
clear port-security interface <int>	Clear port-security info

SECURITY TIP



Always monitor port-security violations and review logs regularly to detect unauthorized access attempts early.





CISCO IOS & VLAN SECURITY

DHCP SNOOPING & DAI



DID YOU KNOW?

Attackers can use rogue DHCP servers or ARP spoofing to steal IP addresses, intercept traffic, or launch man-in-the-middle attacks. DHCP Snooping and DAI stop them!



KEY TAKEAWAY

- ✓ DHCP Snooping builds a trusted IP-MAC binding database.
- ✓ DAI validates ARP packets using this database.
- ✓ Together, they stop IP spoofing, ARP spoofing, and MITM attacks.



@manishburdak45

DHCP SNOOPING & DAI

DHCP Snooping and Dynamic ARP Inspection (DAI) are powerful Layer 2 security features that prevent IP address attacks and ARP spoofing in your network.

“

Trust your infrastructure,
verify your neighbors,
secure your network.



”

28
/ 30

1. DHCP SNOOPING OVERVIEW

DHCP Snooping builds a binding database of valid IP-MAC-VLAN from trusted DHCP responses.

- ✓ Allows only trusted DHCP servers
- ✓ Blocks rogue DHCP servers
- ✓ Builds IP-MAC-VLAN binding table
- ✓ Protects against IP address exhaustion and man-in-the-middle attacks



Works at Layer 2 (Data Link Layer)

2. HOW DHCP SNOOPING WORKS



3. DAI (DYNAMIC ARP INSPECTION)

DAI validates ARP packets using the DHCP Snooping binding table.

- ✓ Drops ARP packets with invalid IP-MAC mappings
- ✓ Prevents ARP spoofing, MITM attacks
- ✓ Ensures only valid hosts can talk



DAI = ARP Security Guard

4. BENEFITS



Prevents Rogue DHCP Servers
Stops attackers from assigning fake IP addresses.



Stops ARP Spoofing
Prevents MITM and session hijacking.



Improves Network Integrity
Ensures trusted and accurate Layer 2 communication.



Easy to Manage
Centralized database, automatic learning and validation.

5. DHCP SNOOPING FEATURES

FEATURE	DESCRIPTION
Trusted / Untrusted Ports	Define which ports can send DHCP responses.
DHCP Snooping Database	Stores IP-MAC-VLAN-Port bindings.
Rate Limiting	Limits DHCP packets on untrusted ports.
Option 82	Inserts information for DHCP relay and tracking.
Database Aging	Removes old bindings automatically.

6. DHCP SNOOPING CONFIGURATION (EXAMPLE)

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 10,20,30
Switch(config)# ip dhcp snooping information option
Switch(config)# interface gig0/1
Switch(config-if)# ip dhcp snooping trust
Switch(config)# interface range gig0/2 - 24
Switch(config-if-range)# ip dhcp snooping limit rate 15
Switch(config-if-range)# end
```

! Verify DHCP Snooping

```
Switch# show ip dhcp snooping binding
IP Address    MAC Address    VLAN  Interface Lease(sec)
192.168.10.10  00aa.bbccc.ddee 10    Gi0/2      86120
192.168.20.15  00aa.bbccc.eeff 20    Gi0/5      86215
```

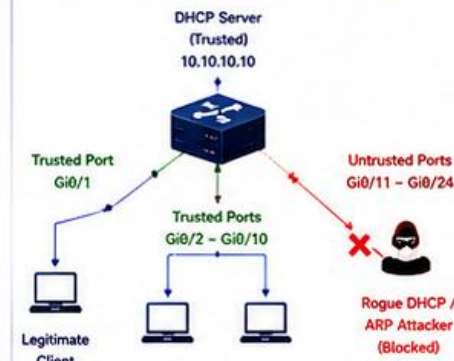
7. DAI CONFIGURATION (EXAMPLE)

```
Switch(config)# ip arp inspection vlan 10,20,30
Switch(config)# interface range gig0/1 - 24
Switch(config-if-range)# ip arp inspection
Switch(config)# interface gig0/1
Switch(config-if)# ip arp inspection trust
Switch(config)# end
```

! Verify DAI

```
Switch# show ip arp inspection statistics
Total Packets      : 24560
Valid Packets      : 24120
Invalid Packets    : 440
Dropped Packets    : 440
```

8. NETWORK SCENARIO



9. COMMON ATTACKS PREVENTED



Rogue DHCP Server Attack
Attacker sets up fake DHCP server.



IP Address Exhaustion
Attacker gives huge IP leases.



ARP Spoofing / Poisoning
Attacker sends fake ARP replies.



Man-in-the-Middle (MITM) Attack
Intercepts and alters traffic.



Bottom Line: DHCP Snooping builds **trust**. DAI enforces it.
Together, they **secure** your network from IP and ARP based attacks.





CISCO IOS & VLAN SECURITY

DHCP SNOOPING & DAI



DID YOU KNOW?

A rogue DHCP server can give attackers IP addresses, default gateways, and DNS settings to steal data or redirect traffic. DAI blocks fake ARP replies that can hijack your network.

- ★ KEY TAKEAWAY**
- ✓ Enable DHCP Snooping on all VLANs.
 - ✓ Trust only uplink ports to DHCP servers.
 - ✓ Enable DAI to stop ARP spoofing.
 - ✓ Always verify and monitor bindings and ARP tables.

@manishburdak45

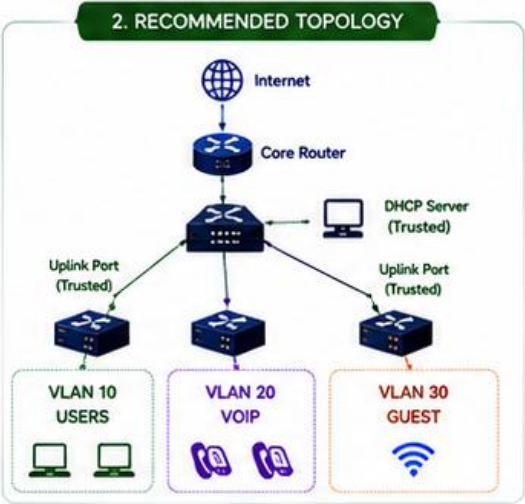
DHCP SNOOPING & DAI BEST PRACTICES

Implementing DHCP Snooping and DAI correctly ensures only **authorized** devices get IP addresses and **prevents** ARP based attacks.

“Layer 2 security is not optional, it's **essential**. Secure your network at the foundation.”



- 1. IMPLEMENTATION FLOW**
- Enable DHCP Snooping globally
 - Define trusted uplink ports
 - Enable DHCP Snooping on VLANs
 - Verify DHCP Snooping bindings
 - Enable DAI on VLANs
 - Verify ARP inspection statistics
 - Monitor and maintain regularly



- 3. SECURITY CHECKLIST**
- ✓ Enable DHCP Snooping on all VLANs
 - ✓ Trust only necessary uplink ports
 - ✓ Verify DHCP Snooping bindings regularly
 - ✓ Enable DAI to prevent ARP spoofing
 - ✓ Use IP Source Guard (Optional)
 - ✓ Log and Monitor security events
 - ✓ Keep switch IOS up to date
 - ✓ Document configuration and policies
- Defense in depth starts at Layer 2 and stops attacks before they start.

4. VERIFICATION COMMANDS

PURPOSE	COMMAND
DHCP Snooping Binding Table	show ip dhcp snooping binding
DHCP Snooping Statistics	show ip dhcp snooping
DHCP Snooping Trusted Ports	show ip dhcp snooping interface
DAI Statistics	show ip arp inspection statistics
ARP Table	show ip arp inspection
Logging	show logging include DHCP ARP

5. DHCP SNOOPING CONFIGURATION (EXAMPLE)

```
Switch# configure terminal
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 10,20,30
Switch(config)# ip dhcp snooping information option
Switch(config)# interface gi0/1
Switch(config-if)# ip dhcp snooping trust
Switch(config-if)# exit
Switch(config)# interface range gi0/2 - 24
Switch(config-if-range)# ip dhcp snooping limit rate 15
Switch(config-if-range)# end

! Verify DHCP Snooping
Switch# show ip dhcp snooping binding
IP Address      MAC Address    VLAN  Interface  Lease(sec)
-----
192.168.10.10   08aa.bbccc.ddee  10    Gi0/2      86120
192.168.20.15   08aa.bbccc.eeff  20    Gi0/5      86215
```

6. DAI CONFIGURATION (EXAMPLE)

```
Switch# configure terminal
Switch(config)# ip arp inspection vlan 10,20,30
Switch(config)# interface range gi0/1 - 24
Switch(config-if-range)# ip arp inspection
Switch(config-if-range)# ip arp inspection limit rate 15
Switch(config-if-range)# end

! Verify DAI
Switch# show ip arp inspection statistics
Total Packets      : 24560
Valid Packets      : 24120
Invalid Packets    : 440
Dropped Packets    : 440

Switch# show ip arp inspection vlan 10
Total ARP Entries  : 125
Dynamic Entries    : 110
Static Entries     : 15
Invalid Entries    : 0
```

- 7. MONITORING & ALERTS**
- Enable logging for DHCP/ARP events
 - Send alerts for violations
 - Regularly review logs and statistics
 - Identify rogue devices quickly
 - Take action and update policies
- Monitoring Tip:** Monitoring is not just about detecting attacks, it's about stopping them early and continuously improving security.

- 8. COMMON MISCONFIGURATIONS**
- ✗ Trusting all ports (including access ports)
 - ✗ Not enabling DHCP Snooping on all VLANs
 - ✗ Not enabling DAI
 - ✗ Not verifying bindings and ARP tables
 - ✗ Ignoring logs and alerts
 - ✗ Allowing unlimited DHCP rate on access ports
- Result:** Attackers can use rogue DHCP servers or ARP spoofing to compromise your network.

Bottom Line: DHCP Snooping ensures only **trusted** DHCP servers assign IP addresses. DAI ensures only **valid** ARP packets are allowed. Together, they build a **secure, reliable, and resilient** network.





CISCO IOS & VLAN SECURITY COMPLETE SUMMARY



DID YOU KNOW?

Proper segmentation, access control, and monitoring can stop 90%+ of network attacks before they cause damage.



KEY TAKEAWAYS

- ✓ Use VLANs to segment networks.
- ✓ Control access with ACLs and port security.
- ✓ Use DHCP Snooping and DAI to prevent IP and ARP attacks.
- ✓ Monitor and log everything.
- ✓ Follow security best practices consistently.



@manishburdak45

COMPLETE VLAN SECURITY SUMMARY

A layered security approach using VLANs, access controls, and threat prevention technologies helps you build a **secure**, **reliable**, and **resilient** network.

1. SECURITY LAYERS

- Network Segmentation (VLANs)
- Access Control (ACLs, Port Security)
- Threat Prevention (DHCP Snooping, DAI)
- Monitoring & Logging (Syslog, SNMP, NetFlow)
- Management & Policies (AAA, SSH, Hardening)

Defense in depth = Strong security

2. RECOMMENDED ARCHITECTURE



3. ESSENTIAL SECURITY FEATURES

- ✓ VLANs for segmentation
- ✓ ACLs to control traffic
- ✓ Port Security to restrict MACs
- ✓ DHCP Snooping to block rogue DHCP
- ✓ DAI to stop ARP spoofing
- ✓ IP Source Guard (Optional)
- ✓ BPDU Guard & Root Guard
- ✓ Disable unused ports & services
- ✓ Regular monitoring and logging
- ✓ Keep IOS updated

Use only what you need, but use it well.

4. COMMON GOALS

- Prevent unauthorized access
- Stop IP and ARP based attacks
- Protect network resources
- Ensure availability and integrity
- Detect and respond quickly
- Comply with security policies

Security is a continuous journey, not a one-time task.

5. BEST PRACTICES

- ✓ Plan your VLANs and IP scheme
- ✓ Apply least privilege access
- ✓ Enable DHCP Snooping and DAI
- ✓ Use strong passwords & AAA
- ✓ Log, monitor, and alert
- ✓ Regularly audit your network
- ✓ Backup configurations
- ✓ Train your team

Good habits today, strong security tomorrow.

6. VLAN CONFIG EXAMPLE

```
vlan 10
 name USERS
vlan 20
 name VOICE
vlan 30
 name SERVERS
vlan 40
 name GUEST
!
interface range fa0/1 - 10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
```

7. DHCP SNOOPING EXAMPLE

```
ip dhcp snooping
ip dhcp snooping vlan 10,20,30,40
interface gi0/1
 ip dhcp snooping trust
interface range gi0/2 - 24
 ip dhcp snooping limit rate 15
```

DHCP Snooping Database (Example)

IP Address	MAC Address	VLAN	Interface	Lease(sec)
192.168.10.10	00aa.bbccc.ddee	10	Gi0/2	86120
192.168.20.15	00aa.bbccc.eeff	20	Gi0/5	86215

8. DAI (ARP INSPECTION) EXAMPLE

```
ip arp inspection vlan 10,20,30,40
interface range gi0/1 - 24
 ip arp inspection trust
interface range gi0/2 - 24
 ip arp inspection limit rate 15
```

DAI Statistics (Example)

Total Packets	: 24560
Valid Packets	: 24120
Invalid Packets	: 440
Dropped Packets	: 440

9. PORT SECURITY EXAMPLE

```
interface fa0/1
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict
```

Port Security Status (Example)

Port	Mode	Max	Secure MACs	Violation
Fa0/1	Secure	2	1	Restrict

10. MONITORING & LOGGING

- Enable logging
logging host 192.168.30.100
- Monitor with SNMP
snmp-server community public R0
- NetFlow for traffic analysis
ip flow-export destination 192.168.30.100
- Regularly review logs & alerts

Example Log (Syslog)

```
*Mar 1 10:15:23: %DHCP_SNOOPING-S-DHCP_SNOOPING_BINDING_ADD:
Binding addition. Interface: Gi0/2, VLAN: 10,
IP: 192.168.10.10, MAC: 00aa.bbccc.ddee
```

“

Security is not a product,
it's a process. Build in **layers**,
monitor always, **stay ahead**.



”

FINAL
SLIDE

30
/ 30



The Bottom Line: Combine VLANs, access controls, DHCP Snooping, and DAI with continuous monitoring to build a **secure**, **scalable**, and **high-performing** network. Security is everyone's responsibility!



Secure Today



Protect Always



Stay Ahead